

Qatar-1b

R-2822

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_250811 · created 2026-09-05T02:38:54+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460898.8794 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.056
Transit depth (Rp/Rs)^2	3.66 +/- 2.13 [%]
Orbital Inclination (inc)	88.82 +/- 2.79
Airmass coefficient 1 (a1)	1.78 +/- 0.15
Airmass coefficient 2 (a2)	-0.36 +/- 0.12
Scatter in the residuals of the lightcurve fit is	9.13 %
Best Comparison Star	#2 - [355, 440]
Optimal Aperture	4.45
Optimal Annulus	11.08
Transit Duration (day)	0.0832 +/- 0.0072

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.191 ± 0.056 vs 0.1463 (deviation 0.8σ)
Duration fit vs published	0.0832 d vs 0.0692 d (deviation 1.95σ)
Mid-time O–C	-5.8 min
Depth SNR	3.4
Residual scatter	9.13 %

Light curve

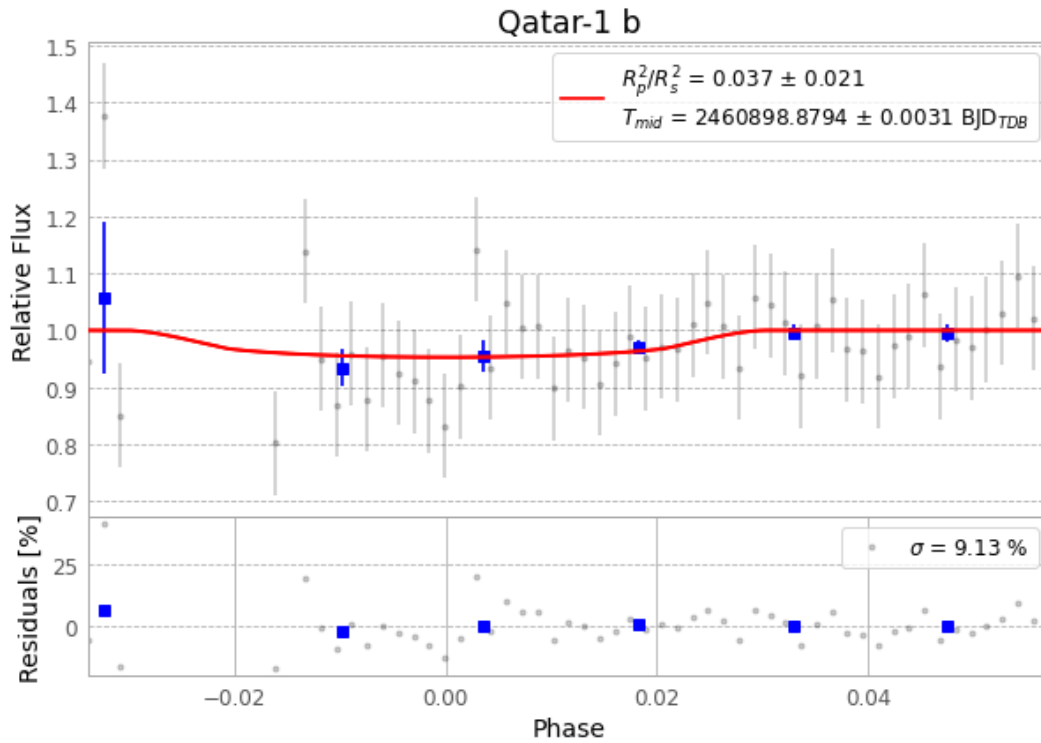


Figure 1 — FinalLightCurve_Qatar-1 b_2025-08-11.png (SHA-256 cd1e1a4dae7cbf39...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [342, 273], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 991eeaeef8193ece1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), Qatar-1250811075416.FITS → Qatar-1250811110015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-250811.log (9f710518d628...), FinalLightCurve_Qatar-1 b_2025-08-11.png (cd1e1a4dae7c...), FinalLightCurve_Qatar-1 b_2025-08-11.csv (04099a736bdf...)

Compute resources

Wall time 376.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 02:38Z.